

ALTOS® Lite Loose Tube, Gel-Free, Single-Jacket, Single-Armored Cable 48 F, TXF™ optical fiber, Single-mode (OS2)

CORNING

Part Number:
048DUC-T4101D20

Corning ALTOS® Lite gel-free, single-jacket, single-armored cables are designed for campus backbones in direct-buried installations. The loose tube design provides stable and highly reliable transmission parameters for a variety of voice, data, video and imaging applications. These cables also provide high-fiber density within a given cable diameter while allowing flexibility to suit many system configurations. The single armored construction provides additional crush and rodent protection with a high-strength ripcord under the armor for easy stripping. Gel-free means the cables are fully waterblocked using craft-friendly, water-swappable materials which make cable access simple and require no clean up. The flexible, craft-friendly buffer tubes are easy to route in closures, and the SZ-stranded, loose tube design isolates fibers from installation and environmental rigors while allowing easy mid-span access. These cables have a medium density polyethylene jacket that is rugged, durable and easy to strip.

Features and Benefits

Gel-free waterblocking technology

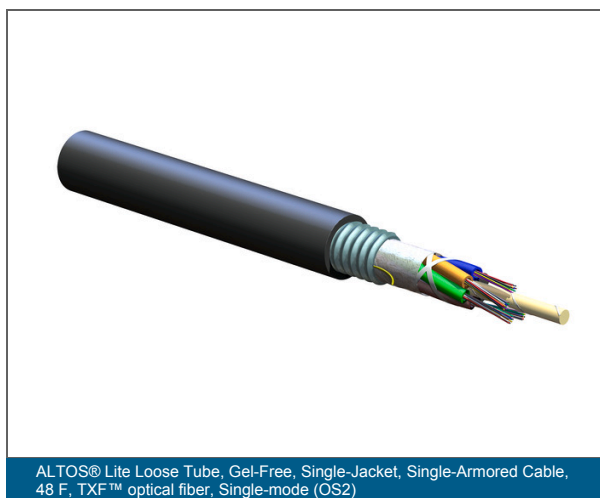
Craft-friendly cable preparation

Polyethylene jacket

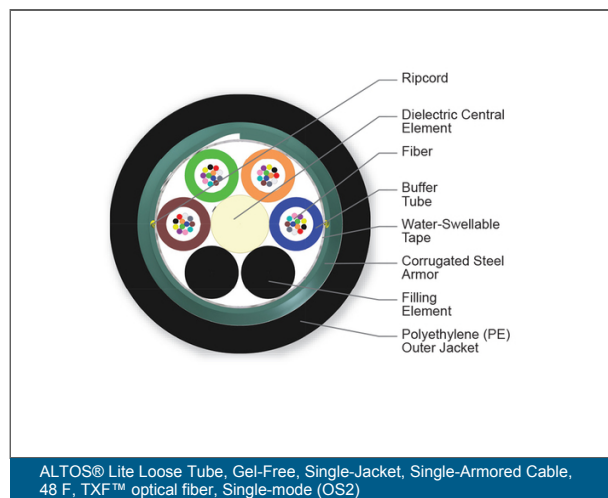
Rugged, durable and easy to strip (while providing superior protection against UV radiation, fungus, abrasion and other environmental factors)

Corrugated steel tape armor

Provides rodent resistance for direct-buried applications



ALTOS® Lite Loose Tube, Gel-Free, Single-Jacket, Single-Armored Cable, 48 F, TXF™ optical fiber, Single-mode (OS2)



ALTOS® Lite Loose Tube, Gel-Free, Single-Jacket, Single-Armored Cable, 48 F, TXF™ optical fiber, Single-mode (OS2)

ALTOS® Lite Loose Tube, Gel-Free, Single-Jacket, Single-Armored Cable 48 F, TXF™ optical fiber, Single-mode (OS2)

CORNING

Specifications

General Specifications

Installation Methods	Aerial, Direct Buried, Duct
Cable Type	Loose Tube
Environment	Outdoor
Product Type	Steel Tape Armor
Fiber Category	TXF®
Cable geometry	Round

Standards

RoHS	Free of hazardous substances according to RoHS 2011/65/EU
Common Installations	Outdoor lashed aerial, duct and direct-buried, indoor when installed according to National Electrical Code® (NEC®) Article 770
Design and Test Criteria	ANSI/ICEA S-87-640

Environmental Conditions

Notes	Corning recommends storing cable in a proper temperature environment prior to installation to allow the cable temperature to meet installation temperature range specifications for best installation results.
Temperature Range, Installation	-30 °C to 70 °C (-22 °F to 158 °F)
Temperature Range, Operation	-40 °C to 70 °C (-40 °F to 158 °F)
Temperature Range, Storage	-40 °C to 70 °C (-40 °F to 158 °F)

Cable Design

Central Element	Dielectric
Fiber Count	48
Number of Ripcords	2
Outer Jacket Color	Black

ALTOS® Lite Loose Tube, Gel-Free, Single-Jacket, Single-Armored Cable 48 F, TXF™ optical fiber, Single-mode (OS2)

CORNING

Cable Design

Outer Jacket Material	Polyethylene (PE)
Tensile Strength Elements and/or Armoring - Layer 1	Corrugated steel tape armor
Buffer Tube Color	Blue, Orange, Green, Brown
Buffer Tube Diameter	2.5 mm (0.1 in)
Number of Active Tubes	4
Number of Filling Elements	2
Number of Tube Positions	6
Tape	Water-swellable
Fiber Coloring	Blue, Orange, Green, Brown, Slate, White, Red, Black, Yellow, Violet, Rose, Aqua
Fibers per Tube	12
Color Code Standards	Telcordia

Mechanical Specifications

Max. Tensile Strength, Long-Term	890 N (200.08 lbf)
Max. Tensile Strength, Short-Term	2700 N (606.98 lbf)
Nominal Outer Diameter	12.1 mm (0.48 in)
Min. Bend Diameter Installation	364 mm (14.33 in)
Min. Bend Diameter Operation	242 mm (9.53 in)

Optical Characteristics

Fiber Code	D
Performance Option Code	01
Fiber Category	OS2
Fiber Type	Single-mode (OS2) / 250 µm
Fiber Name	TXF®
Maximum Attenuation	'- / - / 0.20 dB/km
Wavelengths	1310 nm / 1383 nm / 1550 nm
Fiber Compliance	ITU-T G.654.E

**ALTOS® Lite Loose Tube, Gel-Free, Single-Jacket,
Single-Armored Cable 48 F, TXF™ optical fiber,
Single-mode (OS2)**



Dimensions	
Cable Weight	129 kg/km (86.68 lb/1000 ft)



Corning Optical Communications LLC • 4200 Corning Place • Charlotte, NC • 28216 • United States
800-743-2675 • FAX: 828-325-5060 • International: +1-828-901-5000 • www.corning.com/opcomm

A complete listing of the trademarks of Corning Optical Communications is available at www.corning.com/opcomm/trademarks. All other trademarks are the properties of their respective owners. Corning Optical Communications is ISO 9001 certified. © 2026 Corning Optical Communications. All rights reserved.